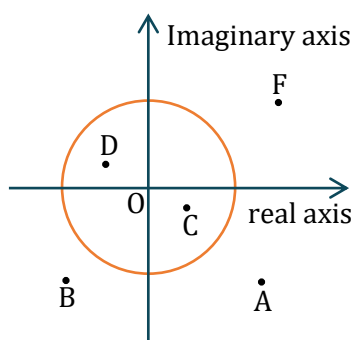


EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- If $z + z^3 = 0$ then which of the following must be true on the complex plane?
(A) $\operatorname{Re}(z) < 0$ (B) $\operatorname{Re}(z) = 0$ (C) $\operatorname{Im}(z) = 0$ (D) $z^4 = 1$
MCN161
- Number of integral values of n for which the quantity $(n + i)^4$ where $i^2 = -1$, is an integer is
(A) 1 (B) 2 (C) 3 (D) 4
MCN162
- Let $i = \sqrt{-1}$. The product of the real part of the roots of $z^2 - z = 5 - 5i$ is
(A) -25 (B) -6 (C) -5 (D) 25
MCN163
- There is only one way to choose real numbers M and N such that when the polynomial $5x^4 + 4x^3 + 3x^2 + Mx + N$ is divided by the polynomial $x^2 + 1$, the remainder is 0. If M and N assume these unique values, then $M - N$ is
(A) -6 (B) -2 (C) 6 (D) 2
MCN164
- Number of complex numbers z satisfying $z^3 = \bar{z}$ is
(A) 1 (B) 2 (C) 4 (D) 5
MCN165
- Let $z = 9 + bi$ where b is non-zero real and $i^2 = -1$. If the imaginary part of z^2 and z^3 are equal, then b^2 equals
(A) 261 (B) 225 (C) 125 (D) 361
MCN166
- The diagram shows several numbers in the complex plane. The circle is the unit circle centered at the origin. One of these numbers is the reciprocal of F , which is



- (A) A (B) B (C) C (D) D

MCN167

- If $z = x + iy$ & $\omega = \frac{1 - iz}{z - i}$ then $|\omega| = 1$ implies that, in the complex plane
(A) z lies on the imaginary axis (B) z lies on the real axis
(C) z lies on the unit circle (D) none

MCN168

9. On the complex plane locus of a point z satisfying the inequality $2 \leq |z - 1| < 3$ denotes
 (A) region between the concentric circles of radii 3 and 1 centered at $(1, 0)$
 (B) region between the concentric circles of radii 3 and 2 centered at $(1, 0)$ excluding the inner and outer boundaries.
 (C) region between the concentric circles of radii 3 and 2 centered at $(1, 0)$ including the inner and outer boundaries.
 (D) region between the concentric circles of radii 3 and 2 centered at $(1, 0)$ including the inner boundary and excluding the outer boundary.

MCN169

10. The complex number z satisfies $z + |z| = 2 + 8i$. The value of $|z|$ is
 (A) 10 (B) 13 (C) 17 (D) 23

MCN170

11. Let $Z_1 = (8 + i)\sin \theta + (7 + 4i)\cos \theta$ and $Z_2 = (1 + 8i)\sin \theta + (4 + 7i)\cos \theta$ are two complex numbers. If $Z_1 \cdot Z_2 = a + ib$ where $a, b \in R$ then the largest value of $(a + b) \forall \theta \in R$, is
 (A) 75 (B) 100 (C) 125 (D) 130

MCN171

12. The locus of z , for $\arg z = -\frac{\pi}{3}$ is
 (A) same as the locus of z for $\arg z = \frac{2\pi}{3}$
 (B) same as the locus of z for $\arg z = \frac{\pi}{3}$
 (C) the part of the straight line $\sqrt{3}x + y = 0$ with $(y < 0, x > 0)$
 (D) the part of the straight line $\sqrt{3}x + y = 0$ with $(y > 0, x < 0)$

MCN172

13. If z_1 & $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin & if $\frac{\operatorname{Im} z_1}{\operatorname{Re} z_1} = \sqrt{2} - 1$ then the value of n is equal to :
 (A) 8 (B) 12 (C) 16 (D) 24

MCN173

14. For $z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ which of the following holds good?
 (A) $\sum |Z_1|^2 = \frac{3}{2}$ (B) $|Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$
 (C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$ (D) $|Z_1|^4 + |Z_2|^4 = |Z_3|^8$

MCN174

15. If z is a complex number satisfying the equation $|z + i| + |z - i| = 8$, on the complex plane then maximum value of $|z|$ is
 (A) 2 (B) 4 (C) 6 (D) 8

MCN175

16. Let Z be a complex number satisfying the equation $(Z^3 + 3)^2 = -16$ then $|Z|$ has the value equal to
 (A) $5^{1/2}$ (B) $5^{1/3}$ (C) $5^{2/3}$ (D) 5

MCN176

17. If z_1, z_2, z_3 are 3 distinct complex numbers such that $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$, then the value of $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ equals
 (A) 0 (B) 3 (C) 4 (D) 5

MCN177

18. Consider two complex numbers α and β as $\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2$, where $a, b \in R$ and $\beta = \frac{z-1}{z+1}$, $z \neq \pm 1$ where $|z| = 1$, then
 (A) Both α and β are purely real
 (B) Both α and β are purely imaginary
 (C) α is purely real and β is purely imaginary
 (D) β is purely real and α is purely imaginary

MCN178

19. Let Z is complex satisfying the equation $z^2 - (3+i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root. The additive inverse of non real root, is
 (A) $1-i$ (B) $1+i$ (C) $-1-i$ (D) -2

MCN179

20. The minimum value of $|z-1+2i| + |4i-3-z|$ is
 (A) $\sqrt{5}$ (B) 5 (C) $2\sqrt{13}$ (D) $\sqrt{15}$

MCN180

21. Identify the incorrect statement.
 (A) no non zero complex number z satisfies the equation, $\bar{z} = -4z$
 (B) $\bar{z} = z$ implies that z is purely real
 (C) If z is purely imaginary, then $\bar{z} + z = 0$.
 (D) if z_1, z_2 are the roots of the quadratic equation $az^2 + bz + c = 0$ such that $\text{Im}(z_1 z_2) \neq 0$ then a, b, c must be real numbers.

MCN181

22. $z_1 = \frac{a}{1-i}$; $z_2 = \frac{b}{2+i}$; $z_3 = a-bi$ for $a, b \in R$ if $z_1 - z_2 = 1$ then the centroid of the triangle formed by the points z_1, z_2, z_3 in the argand's plane is given by
 (A) $\frac{1}{9}(1+7i)$ (B) $\frac{1}{3}(1+7i)$ (C) $\frac{1}{3}(1-3i)$ (D) $\frac{1}{9}(1-3i)$

MCN182

23. Number of complex numbers z such that $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ is
 (A) 4 (B) 6 (C) 8 (D) more than 8

MCN183

24. If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{2}{z}$, then the locus traced by ' ω ' in the complex plane is

(A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

MCN184

25. A particle starts from a point $z_0 = 1 + i$, where $i = \sqrt{-1}$. It moves horizontally away from origin by 2 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 particle moves $\sqrt{5}$ units in the direction of $2\hat{i} + \hat{j}$ and then it moves through an angle of $\operatorname{cosec}^{-1}\sqrt{2}$ in anticlockwise direction of a circle with centre at origin to reach a point z_2 . The $\arg z_2$ is given by

(A) $\sec^{-1}2$ (B) $\cot^{-1}0$ (C) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$ (D) $\cos^{-1}\left(\frac{-1}{2}\right)$

MCN185

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If z is a complex number which simultaneously satisfies the equations $3|z-12| = 5|z-8i|$ and $|z-4| = |z-8|$ then the $\text{Im}(z)$ can be
 (A) 15 (B) 16 (C) 17 (D) 8

MCN186

2. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$. Which of the following statement(s) is/are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

(B) $\sum_{k=1}^3 \text{Amp}(z_k) = \frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

MCN187

3. The loci of a point $P(z)$ in the complex plane satisfying the $\left|z + \frac{1}{z}\right| = 2$ are two circles C_1 and C_2 .

These circles

(A) have centres on real axis.

(B) cut each other orthogonally.

(C) are congruent

(D) have exactly two common tangents.

MCN188

4. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha, z \neq \beta$. Which of the following relation(s) hold good?

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

(B) $\exists$ a positive real number ' t ' such that $z = (1-t)\alpha + t\beta$

(C) $\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$

(D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

MCN189

5. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where ' O ' is the origin can be written as

(A) $\text{Im}\left(\frac{z-\alpha}{\alpha}\right) = 0$ (B) $\text{Re}\left(\frac{z-\alpha}{\alpha}\right) = 0$ (C) $\text{Re}(\bar{\alpha}z) = 0$ (D) $\bar{\alpha}z + \alpha\bar{z} - 2|\alpha|^2 = 0$

MCN190

6. If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$ are the imaginary n^{th} roots of unity then the product $\prod_{r=1}^{n-1} (i - \alpha_r)$ (where $i = \sqrt{-1}$) can take the value equal to
 (A) 0 (B) 1 (C) i (D) $(1 + i)$ MCN191
7. If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be
 (A) $\cot \frac{\pi}{8}$ (B) $\sec \pi$ (C) $\tan \frac{\pi}{12}$ (D) $\tan \frac{5\pi}{12}$ MCN192
8. Locus of all points z in argand plane which satisfy $|z^2 + 1| = |z^2 - 1|$ is -
 (A) same as $\text{Re}(z) = 0$ (B) same as $\text{Re}(z^2) = 0$
 (C) a pair of straight lines (D) a circle with unit radius MCN193
9. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ ($a, b, c, d \in R$) has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$.
 (A) $b = 51$ (B) $a = -6$ (C) $c = -70$ (D) $d = 170$ MCN194
10. The quadratic equation $z^2 + (p + ip')z + q + iq' = 0$; where p, p', q, q' are all real.
 (A) if the equation has one real root then $q'^2 - pp'q' + qp'^2 = 0$.
 (B) if the equation has two equal roots then $pp' = 2q'$.
 (C) if the equation has two equal roots then $p^2 - p'^2 = 4q$
 (D) if the equation has one real root then $p'^2 - pp'q' + q'^2 = 0$. MCN195
11. The value of $i^n + i^{-n}$, for $i = \sqrt{-1}$ and $n \in I$ is :
 (A) $\frac{2^n}{(1-i)^{2n}} + \frac{(1+i)^{2n}}{2^n}$ (B) $\frac{(1+i)^{2n}}{2^n} + \frac{(1-i)^{2n}}{2^n}$
 (C) $\frac{(1+i)^{2n}}{2^n} + \frac{2^n}{(1-i)^{2n}}$ (D) $\frac{2^n}{(1+i)^{2n}} + \frac{2^n}{(1-i)^{2n}}$ MCN196
12. If $\text{amp}(z_1 z_2) = 0$ and $|z_1| = |z_2| = 1$, then
 (A) $z_1 + z_2 = 0$ (B) $z_1 z_2 = 1$ (C) $z_1 = \bar{z}_2$ (D) $z_1 = z_2$ MCN197
13. Let z_1 and z_2 are two complex numbers such that $(1 - i)z_1 = 2z_2$ and $\arg(z_1 z_2) = \frac{\pi}{2}$, then $\arg(z_2)$ is equal to
 (A) $3\pi/8$ (B) $\pi/8$ (C) $5\pi/8$ (D) $-7\pi/8$ MCN198
14. a, b, c are real numbers in the polynomial, $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$. If two roots of the equation $P(z) = 0$ are 2 and i . Then which of the following are true.
 (A) $a = -\frac{11}{2}$ (B) $b = 5$ (C) $c = -\frac{11}{2}$ (D) $a = -11$ MCN199

15. If $Z = \frac{(1+i)(1+2i)(1+3i)\dots(1+ni)}{(1-i)(2-i)(3-i)\dots(n-i)}$, $n \in N$ then principal argument of Z can be
- (A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π

MCN200

COMPREHENSION TYPE QUESTIONS

Paragraph for question nos. 16 to 18

Consider a complex number $w = \frac{z-i}{2z+1}$, where $z = x + iy$ and $x, y \in R$.

16. If the complex number w is purely imaginary then locus of z is -
- (A) a straight line
(B) a circle with centre $\left(-\frac{1}{4}, \frac{1}{2}\right)$ and radius $\frac{\sqrt{5}}{4}$.
(C) a circle with centre $\left(\frac{1}{4}, -\frac{1}{2}\right)$ and passing through origin.
(D) neither a circle nor a straight line.

MCN201

17. If the complex number w is purely real then locus of z is
- (A) a straight line passing through origin
(B) a straight line with gradient 3 and y intercept (-1)
(C) a straight line with gradient 2 and y intercept 1.
(D) none

MCN202

18. If $|w| = 1$ then the locus of $P(z)$ is
- (A) a point circle (B) an imaginary circle
(C) a real circle (D) not a circle.

MCN203

Paragraph for question nos. 19 to 20

$ABCD$ is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy $BD = 2AC$. Let the points D and M represent complex numbers $1 + i$ and $2 - i$ respectively.

If θ is arbitrary real, then $z = re^{i\theta}$, $R_1 \leq r \leq R_2$ lies in annular region formed by concentric circles $|z| = R_1$, $|z| = R_2$.

19. A possible representation of point A is
- (A) $3 - \frac{i}{2}$ (B) $1 - \frac{3i}{2}$ (C) $1 + \frac{3}{2}i$ (D) $3 - \frac{3}{2}i$

MCN204

20. If z is any point on segment DM then $w = e^{iz}$ lies in annular region formed by concentric circles.
- (A) $|w|_{\min} = 1$, $|w|_{\max} = 2$ (B) $|w|_{\min} = \frac{1}{e}$, $|w|_{\max} = e$
(C) $|w|_{\min} = \frac{1}{e^2}$, $|w|_{\max} = e^2$ (D) $|w|_{\min} = \frac{1}{2}$, $|w|_{\max} = e$

MCN205

Paragraph for question nos. 21 to 23

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z+1| \leq 2 + \operatorname{Re}(z)\}, B = \{z : |z-1| \geq 1\} \text{ and } C = \left\{z : \left|\frac{z-1}{z+1}\right| \geq 1\right\}$$

21. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is MCN206
22. The area of region bounded by $A \cap B \cap C$ is $2\sqrt{\lambda}$ then λ is MCN207
23. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude can be MCN208

MATCHING LIST TYPE QUESTION

24. Match the List

List - I

(Complex number Z)

(A) $z = \frac{(1+i)^5 (1+\sqrt{3}i)^2}{-2i(-\sqrt{3}+i)}$

(B) $z = \sin \frac{6\pi}{5} + i \left(1 + \cos \frac{6\pi}{5}\right)$ is

(C) $z = 1 + \cos\left(\frac{11\pi}{9}\right) + i \sin\left(\frac{11\pi}{9}\right)$

(D) $z = \sin x \sin(x-60) \sin(x+60)$

where $x \in \left(0, \frac{\pi}{3}\right)$ and $x \in R$

List - II

(Principal argument of Z)

(P) π

(Q) $-\frac{7\pi}{18}$

(R) $\frac{9\pi}{10}$

(S) $-\frac{5\pi}{12}$

(T) 0

(A) $(A) \rightarrow (P), (B) \rightarrow (R), (C) \rightarrow (S), (D) \rightarrow (Q)$

(B) $(A) \rightarrow (S), (B) \rightarrow (R), (C) \rightarrow (Q), (D) \rightarrow (P)$

(C) $(A) \rightarrow (R), (B) \rightarrow (Q), (C) \rightarrow (S), (D) \rightarrow (P)$

(D) $(A) \rightarrow (Q), (B) \rightarrow (S), (C) \rightarrow (R), (D) \rightarrow (P)$

MCN209

MATRIX MATCH TYPE QUESTION

25. Match the equation in z , in **Column-I** with the corresponding values of $\arg(z)$ in **Column-II**.

Column-I

(equations in z)

(A) $z^2 - z + 1 = 0$

(B) $z^2 + z + 1 = 0$

(C) $2z^2 + 1 + i\sqrt{3} = 0$

(D) $2z^2 + 1 - i\sqrt{3} = 0$

Column-II

(principal value of $\arg(z)$)

(P) $-2\pi/3$

(Q) $-\pi/3$

(R) $\pi/3$

(S) $2\pi/3$

MCN210

EXERCISE (S)

1. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer then find the value of $\frac{N}{2}$.
MCN211
2. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg zw = \pi$. If $\operatorname{Re}(z) < 0$ and principal $\arg z = \frac{a\pi}{b}$ then find the value of $a + b$. (where a & b are co-prime natural numbers)
MCN212
3. If $x = 9^{1/3} 9^{1/9} 9^{1/27} \dots \infty$, $y = 4^{1/3} 4^{-1/9} 4^{1/27} \dots \infty$, and $z = \sum_{r=1}^{\infty} (1+i)^{-r}$ and principal argument of $P = (x + yz)$ is $-\tan^{-1}\left(\frac{\sqrt{a}}{b}\right)$ then determine $a^2 + b^2$. (where a & b are co-prime natural numbers)
MCN213
4. $z_1, z_2 \in \mathbb{C}$ and $z_1^2 + z_2^2 \in \mathbb{R}$, $z_1(z_1^2 - 3z_2^2) = 2$, $z_2(3z_1^2 - z_2^2) = 11$. If $z_1^2 + z_2^2 = \lambda$ then determine λ^2
MCN214
5. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in \mathbb{C}$ (where $\mathbb{C}$ is the set of complex numbers), then find product of maximum and minimum value of $|w|$.
MCN215
6. A function ' f ' is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex number z , where α and γ are complex numbers if $f(1)$ and $f(i)$ are both real and the smallest possible values of $|\alpha| + |\gamma|$ is p then determine p .
MCN216
7. If z and ω are two non-zero complex numbers such that $|z\omega| = 1$, and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then find the value of $\frac{5i\bar{z}\omega}{4}$
MCN217
8. If $a_1, a_2, a_3, \dots, a_n, A_1, A_2, A_3, \dots, A_n, k$ are all real numbers and number of imaginary roots of the equation $\frac{A_1^2}{x - a_1} + \frac{A_2^2}{x - a_2} + \dots + \frac{A_n^2}{x - a_n} = k$ is α . Then find the value of $\frac{\alpha + 15}{10}$.
MCN218
9. If a variable circle S touches $S_1 : |z - z_1| = 7$ internally and $S_2 : |z - z_2| = 4$ externally while the curves S_1 & S_2 touch internally to each other, ($z_1 \neq z_2$). If the eccentricity of the locus of the centre of the curve S is ' e ' find the value of e .
MCN219
10. Area of the region formed by $|z| \leq 4$ & $-\frac{\pi}{2} \leq \arg z \leq \frac{\pi}{3}$ on the Argand diagram is expressed in the form of $\frac{a\pi}{b}$. Then find the value of $\frac{a}{b}$ (where a & b are co-prime natural number)
MCN220

EXERCISE - JEE (Main) PYQ

1. Let $z \in \mathbb{C}$ be such that $|z| < 1$. If $\omega = \frac{5+3z}{5(1-z)}$, then - [JEE (Main) 2019]
 (1) $5\text{Im}(\omega) < 1$ (2) $4\text{Im}(\omega) > 5$ (3) $5\text{Re}(\omega) > 1$ (4) $5\text{Re}(\omega) > 4$
MCN035
2. If $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x+iy$, then the point (x, y) lies on a : [JEE (Main) 2020]
 (1) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$ (2) circle whose diameter is $\frac{\sqrt{5}}{2}$
 (3) straight line whose slope is $\frac{3}{2}$ (4) straight line whose slope is $-\frac{2}{3}$
MCN036
3. The value of $\left(\frac{1 + \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9}}{1 + \sin \frac{2\pi}{9} - i \cos \frac{2\pi}{9}}\right)^3$ is : [JEE (Main) 2020]
 (1) $\frac{1}{2}(\sqrt{3} - i)$ (2) $-\frac{1}{2}(\sqrt{3} - i)$ (3) $-\frac{1}{2}(1 - i\sqrt{3})$ (4) $\frac{1}{2}(1 - i\sqrt{3})$
MCN037
4. If a and b are real numbers such that $(2 + \alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then $a + b$ is equal to : [JEE (Main) 2020]
 (1) 57 (2) 33 (3) 24 (4) 9
MCN038
5. The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is : [JEE (Main) 2020]
 (1) $2^{15}i$ (2) -2^{15} (3) $-2^{15}i$ (4) 6^5
MCN039
6. Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to. [JEE (Main) 2021]
MCN040
7. Let the lines $(2-i)z = (2+i)\bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C . If the line $iz + \bar{z} + 1 + i = 0$ is tangent to this circle C , then its radius is: [JEE (Main) 2021]
 (1) $\frac{3}{\sqrt{2}}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $3\sqrt{2}$ (4) $\frac{3}{2\sqrt{2}}$
MCN041

8. The least value of $|z|$ where z is complex number which satisfies the inequality

$$\exp\left(\frac{(|z|+3)(|z|-1)}{|z|+1}\log_e 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7}+9i|, i=\sqrt{-1}, \text{ is equal to :}$$

[JEE (Main) 2021]

- (1) 3 (2) $\sqrt{5}$ (3) 2 (4) 8

MCN042

9. Let S_1, S_2 and S_3 be three sets defined as

$$S_1 = \{z \in \mathbb{C} : |z-1| \leq \sqrt{2}\}$$

$$S_2 = \{z \in \mathbb{C} : \operatorname{Re}((1-i)z) \geq 1\}$$

$$S_3 = \{z \in \mathbb{C} : \operatorname{Im}(z) \leq 1\}$$

Then the set $S_1 \cap S_2 \cap S_3$

[JEE (Main) 2021]

- (1) is a singleton (2) has exactly two elements
(3) has infinitely many elements (4) has exactly three elements

MCN043

10. If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then

$\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ is : (Here $\arg(z)$ denotes the principal argument of complex number z)

[JEE (Main) 2021]

- (1) $\frac{\pi}{4}$ (2) $-\frac{3\pi}{4}$ (3) $-\frac{\pi}{4}$ (4) $\frac{3\pi}{4}$

MCN044

11. Let $\mathbb{C}$ be the set of all complex numbers. Let

$$S_1 = \{z \in \mathbb{C} : |z-2| \leq 1\} \text{ and}$$

$$S_2 = \{z \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \geq 4\}.$$

Then, the maximum value of $\left|z - \frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to :

[JEE (Main) 2021]

- (1) $\frac{3+2\sqrt{2}}{4}$ (2) $\frac{5+2\sqrt{2}}{2}$ (3) $\frac{3+2\sqrt{2}}{2}$ (4) $\frac{5+2\sqrt{2}}{4}$

MCN045

12. Let $z = \frac{1-i\sqrt{3}}{2}, i=\sqrt{-1}$ then the value of

$$21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^3 \text{ is.}$$

[JEE (Main) 2021]

MCN046

13. If $(\sqrt{3}+i)^{100} = 2^{99}(p+iq)$, then p and q are roots of the equation :

[JEE (Main) 2021]

- (1) $x^2 - (\sqrt{3}-1)x - \sqrt{3} = 0$ (2) $x^2 + (\sqrt{3}+1)x + \sqrt{3} = 0$
(3) $x^2 + (\sqrt{3}-1)x - \sqrt{3} = 0$ (4) $x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$

MCN047

14. If $S = \left\{ z \in \mathbb{C} : \frac{z-i}{z+2i} \in \mathbb{R} \right\}$, then : [JEE (Main) 2021]

(1) S contains exactly two elements (2) S contains only one element
(3) S is a circle in the complex plane (4) S is a straight line in the complex plane

MCN048

15. If $a_r = \cos \frac{2r\pi}{9} + i \sin \frac{2r\pi}{9}$, $r = 1, 2, 3, \dots, i = \sqrt{-1}$ then the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is equal to

[JEE (Main) 2021]

(1) $a_2 a_6 - a_4 a_8$ (2) a_9 (3) $a_1 a_9 - a_3 a_7$ (4) a_5

MCN049

16. For $z \in \mathbb{C}$ if the minimum value of $(|z - 3\sqrt{2}| + |z - p\sqrt{2}i|)$ is $5\sqrt{2}$, then a value of p is _____

[JEE (Main) 2022]

(1) 3 (2) $\frac{7}{2}$ (3) 4 (4) $\frac{9}{2}$

MCN050

17. Let S be the set of all (α, β) , $\pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i \cos 2\beta, (\alpha, \beta) \in S$. Then

$\sum_{(\alpha, \beta) \in S} \left(i Z_{\alpha\beta} + \frac{1}{i \bar{Z}_{\alpha\beta}} \right)$ is equal to :

[JEE (Main) 2022]

(1) 3 (2) $3i$ (3) 1 (4) $2 - i$

MCN051

18. Let $S = \{z = x + iy : |z - 1 + i| \geq |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of x , for which $w = 2x + iy \in S$ for some $y \in \mathbb{R}$, is

[JEE (Main) 2022]

(1) $\left[-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$ (2) $\left[-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$ (3) $\left[-\sqrt{2}, \frac{1}{2}\right]$ (4) $\left[-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

MCN052

19. Let $S = \{z \in \mathbb{C} : |z - 3| \leq 1 \text{ and } z(4 + 3i) + \bar{z}(4 - 3i) \leq 24\}$. If $\alpha + i\beta$ is the point in S which is closest to $4i$, then $25(\alpha + \beta)$ is equal to ____.

[JEE (Main) 2022]

MCN053

20. Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z (\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3 , then $\arg(z)$ is equal to :

[JEE (Main) 2022]

(1) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$ (2) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$ (3) $\tan^{-1}(3) - \pi$ (4) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

MCN054

21. The number of points of intersection of $|z - (4 + 3i)| = 2$ and $|z| + |z - 4| = 6$, $z \in \mathbb{C}$ is :

[JEE (Main) 2023]

- (1) 0 (2) 1 (3) 2 (4) 3

MCN055

22. Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$. If $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to

[JEE (Main) 2023]

- (1) 21 (2) $\frac{169}{8}$ (3) 19 (4) $\frac{155}{8}$

MCN056

23. Let the complex number $z = x + iy$ be such that $\frac{2z - 3i}{2z + i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to :

[JEE (Main) 2023]

- (1) $\frac{3}{2}$ (2) $\frac{4}{3}$ (3) $\frac{2}{3}$ (4) $\frac{3}{4}$

MCN057

24. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) > \operatorname{Im}(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) < \operatorname{Im}(\bar{a} + z)\}$. Then among the two statements :

[JEE (Main) 2023]

(S1) : If $\operatorname{Re}(A), \operatorname{Im}(A) > 0$, then the set A contains all the real numbers

(S2) : If $\operatorname{Re}(A), \operatorname{Im}(A) < 0$, then the set B contains all the real numbers,

- (1) Only (S1) is true (2) both are false (3) Only (S2) is true (4) Both are true

MCN058

25. Let $z_1 = 2 + 3i$ and $z_2 = 3 + 4i$. The set $S = \{z \in \mathbb{C} : |z - z_1|^2 - |z - z_2|^2 = |z_1 - z_2|^2\}$ represents a

[JEE (Main) 2023]

(1) straight line with sum of its intercepts on the coordinate axes equals 14

(2) hyperbola with the length of the transverse axis 7

(3) straight line with the sum of its intercepts on the coordinate axes equals -18

(4) hyperbola with eccentricity 2

MCN059

26. Let $\alpha = 8 - 14i$, $A = \left\{z \in \mathbb{C} : \frac{\alpha z - \bar{\alpha} \bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ and $B = \{z \in \mathbb{C} : |z + 3i| = 4\}$.

Then $\sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z)$ is equal to ____.

[JEE (Main) 2023]

MCN060

27. The area (in sq. units) of the region $S = \{z \in \mathbb{C} : |z - 1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \operatorname{Im}(z) \geq 0\}$ is

[JEE (Main) 2024]

- (1) $\frac{7\pi}{3}$ (2) $\frac{3\pi}{2}$ (3) $\frac{17\pi}{8}$ (4) $\frac{7\pi}{4}$

MCN221

28. Consider the following two statements :

Statement I : For any two non-zero complex numbers z_1, z_2

$$\left(|z_1| + |z_2| \right) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq 2(|z_1| + |z_2|) \text{ and}$$

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$, then

$$\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = 1.$$

Between the above two statements,

[JEE (Main) 2024]

- (1) both Statement I and Statement II are incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I is correct but Statement II is incorrect.
- (4) both Statement I and Statement II are correct.

MCN222

29. If z_1, z_2 are two distinct complex number such that $\left| \frac{z_1 - 2z_2}{\frac{1}{2} - z_1 \bar{z}_2} \right| = 2$, then

[JEE (Main) 2024]

- (1) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$
- (2) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1.
- (3) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1.
- (4) both z_1 and z_2 lie on the same circle.

MCN223

30. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and

$$(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2) = \frac{125}{16}m. \text{ Then the value of } m \text{ is } \underline{\hspace{2cm}}.$$

[JEE (Main) 2024]

MCN224

31. If the set $R = \{(a, b) ; a + 5b = 42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^m (1 + i^{n!}) = x + iy$, where

$I = \sqrt{-1}$, then the value of $m + x + y$ is :

[JEE (Main) 2024]

- (1) 8 (2) 12 (3) 4 (4) 5

MCN225

32. Let $S = \{z \in \mathbb{C} : |z-1| = 1 \text{ and } (\sqrt{2}-1)(z+\bar{z}) - i(z-\bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that

$$|z_1| = \max_{z \in S} |z| \text{ and } |z_2| = \min_{z \in S} |z|. \text{ Then } |\sqrt{2}z_1 - z_2|^2 \text{ equals :}$$

[JEE (Main) 2024]

- (1) 1 (2) 4 (3) 3 (4) 2

MCN226

33. Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100|\alpha|^2$ is.....

[JEE (Main) 2024]

MCN227

34. Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\text{Im}(\alpha) > \text{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib), i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

[JEE (Main) 2024]

MCN228

35. If α denotes the number of solutions of $|1 - i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)} \right)$,

where $z = \frac{\pi}{4}(1+i)^4 \left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i} \right)$, $i = \sqrt{-1}$, then the distance of the point (α, β) from the

line $4x - 3y = 7$ is _____

[JEE (Main) 2024]

MCN229

EXERCISE - JEE (Advanced) PYQ

1. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when $n =$ [JEE (Advanced) 2013]
 (A) 57 (B) 55 (C) 58 (D) 56

MCN104

2. Let $w = \frac{\sqrt{3}+i}{2}$ and $P = \{w^n : n = 1, 2, 3, \dots\}$. Further $H_1 = \left\{z \in \mathbb{C} : \operatorname{Re} z > \frac{1}{2}\right\}$ and $H_2 = \left\{z \in \mathbb{C} : \operatorname{Re} z < \frac{-1}{2}\right\}$ where $\mathbb{C}$ is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2 =$ [JEE (Advanced) 2013]
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

MCN105

Paragraph for Question 3 and 4

Let $S = S_1 \cap S_2 \cap S_3$, where

$$S_1 = \{z \in \mathbb{C} : |z| < 4\}, S_2 = \left\{z \in \mathbb{C} : \operatorname{Im} \left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i} \right] > 0\right\} \text{ and } S_3 = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}.$$

3. $\min_{z \in S} |1-3i-z| =$ [JEE (Advanced) 2013]
 (A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$ (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$
4. Area of $S =$ [JEE (Advanced) 2013]
 (A) $\frac{10\pi}{3}$ (B) $\frac{20\pi}{3}$ (C) $\frac{16\pi}{3}$ (D) $\frac{32\pi}{3}$

MCN106

MCN107

5. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right)$; $k = 1, 2, \dots, 9$. [JEE (Advanced) 2014]

List-I

P. For each z_k there exists a z_j such that

$$z_k \cdot z_j = 1$$

Q. There exists a $k \in \{1, 2, \dots, 9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.R. $\frac{|1-z_1| |1-z_2| \dots |1-z_9|}{10}$ equalsS. $1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right)$ equals

List-II

1. True

2. False

3. 1

4. 2

Codes :

	P	Q	R	S		P	Q	R	S
(A)	1	2	4	3	(B)	2	1	3	4
(C)	1	2	3	4	(D)	2	1	4	3

MCN108

6. **Column-I** **Column-II**
- (A) In $\mathbb{R}^2$, if the magnitude of the projection vector of the vector $\alpha\hat{i} + \beta\hat{j}$ on $\sqrt{3}\hat{i} + \hat{j}$ is $\sqrt{3}$ and if $\alpha = 2 + \sqrt{3}\beta$, then possible value(s) of $|\alpha|$ is (are) (P) 1
- (B) Let a and b be real numbers such that the function $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$ is differentiable for all $x \in \mathbb{R}$. Then possible value(s) of a is (are) (Q) 2
- (C) Let $\omega \neq 1$ be a complex cube root of unity. If $(3 - 3\omega + 2\omega^2)^{4n+3} + (2 + 3\omega - 3\omega^2)^{4n+3} + (-3 + 2\omega + 3\omega^2)^{4n+3} = 0$ then possible value(s) of n is (are) (R) 3
- (D) Let the harmonic mean of two positive real number a and b be 4, If q is a positive real number such that $a, 5, q, b$ is an arithmetic progression, then the value(s) of $|q - a|$ is (are) (S) 4 (T) 5

[JEE (Advanced) 2015]
MCN109

7. For any integer k , let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i \sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|} \text{ is}$$

[JEE (Advanced) 2015]

MCN110

8. Let $z = \frac{-1 + \sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is

[JEE (Advanced) 2016]
MCN079

9. Let $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S = \left\{ z \in \mathbb{C} : z = \frac{1}{a + ibt}, t \in \mathbb{R}, t \neq 0 \right\}$, where $i = \sqrt{-1}$.

If $z = x + iy$ and $z \in S$, then (x, y) lies on

[JEE (Advanced) 2016]

- (A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a > 0, b \neq 0$
 (B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a < 0, b \neq 0$
 (C) the x -axis for $a \neq 0, b = 0$
 (D) the y -axis for $a = 0, b \neq 0$

MCN080

10. Let a, b, x and y be real numbers such that $a - b = 1$ and $y \neq 0$. If the complex number $z = x + iy$ satisfies $\operatorname{Im}\left(\frac{az+b}{z+1}\right) = y$, then which of the following is(are) possible value(s) of x ?

[JEE (Advanced) 2017]

- (A) $-1 - \sqrt{1 - y^2}$ (B) $1 + \sqrt{1 + y^2}$ (C) $1 - \sqrt{1 + y^2}$ (D) $-1 + \sqrt{1 - y^2}$

MCN081

11. For a non-zero complex number z , let $\arg(z)$ denotes the principal argument with $-\pi < \arg(z) \leq \pi$. Then, which of the following statement(s) is (are) FALSE?

[JEE (Advanced) 2018]

- (A) $\arg(-1 - i) = \frac{\pi}{4}$, where $i = \sqrt{-1}$
 (B) The function $f : \mathbb{R} \rightarrow [-\pi, \pi]$, defined by $f(t) = \arg(-1 + it)$ for all $t \in \mathbb{R}$, is continuous at all points of $\mathbb{R}$, where $i = \sqrt{-1}$
 (C) For any two non-zero complex numbers z_1 and z_2 , $\arg\left(\frac{z_1}{z_2}\right) - \arg(z_1) + \arg(z_2)$ is an integer multiple of 2π
 (D) For any three given distinct complex numbers z_1, z_2 and z_3 , the locus of the point z satisfying the condition $\arg\left(\frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}\right) = \pi$, lies on a straight line

MCN082

12. Let s, t, r be the non-zero complex numbers and L be the set of solutions $z = x + iy$ ($x, y \in \mathbb{R}, i = \sqrt{-1}$) of the equation $sz + t\bar{z} + r = 0$, where $\bar{z} = x - iy$. Then, which of the following statement(s) is (are) TRUE?

[JEE (Advanced) 2018]

- (A) If L has exactly one element, then $|s| \neq |t|$
 (B) If $|s| = |t|$, then L has infinitely many elements
 (C) The number of elements in $L \cap \{z : |z - 1 + i| = 5\}$ is at most 2
 (D) If L has more than one element, then L has infinitely many elements

MCN083

13. Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set $\left\{\frac{1}{|z - 1|} : z \in S\right\}$, then the principal argument of

$$\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i} \text{ is}$$

[JEE (Advanced) 2019]

- (A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

MCN084

14. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a + b\omega + c\omega^2|^2 : a, b, c \text{ distinct non-zero integers}\}$ equals

[JEE (Advanced) 2019]

MCN085

15. Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are TRUE ? [JEE (Advanced) 2020]

- (A) $\left|z + \frac{1}{2}\right| \leq \frac{1}{2}$ for all $z \in S$ (B) $|z| \leq 2$ for all $z \in S$
 (C) $\left|z + \frac{1}{2}\right| \geq \frac{1}{2}$ for all $z \in S$ (D) The set S has exactly four elements

MCN086

16. For a complex number z , let $\text{Re}(z)$ denote the real part of z . Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\text{Re}(z_1) > 0$ and $\text{Re}(z_2) < 0$, is [JEE (Advanced) 2020]

MCN087

17. Let $\theta_1, \theta_2, \dots, \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + \dots + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}e^{i\theta_k}$ for $k = 2, 3, \dots, 10$, where $i = \sqrt{-1}$. Consider the statements P and Q given below : [JEE (Advanced) 2021]

$$P : |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{10} - z_9| + |z_1 - z_{10}| \leq 2\pi$$

$$Q : |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + \dots + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \leq 4\pi$$

Then,

- (A) P is **TRUE** and Q is **FALSE** (B) Q is **TRUE** and P is **FALSE**
 (C) both P and Q are **TRUE** (D) both P and Q are **FALSE**

MCN088

18. For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z + \alpha}{z + \beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 + 5x - 3y + 4 = 0$. Then which of the following statements is (are) **TRUE** ? [JEE (Advanced) 2021]

- (A) $\alpha = -1$ (B) $\alpha\beta = 4$ (C) $\alpha\beta = -4$ (D) $\beta = 4$

MCN089

19. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is _____. [JEE (Advanced) 2022]

MCN090

20. Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\bar{z} - z^2 = i(\bar{z} + z^2)$ is. [JEE (Advanced) 2022]

MCN091

21. Let $\bar{z}$ denote the complex conjugate of a complex number z . If z is a non-zero complex number for which both real and imaginary parts of $(\bar{z})^2 + \frac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of $|z|$?

[JEE (Advanced) 2022]

(A) $\left(\frac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$ (B) $\left(\frac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$ (C) $\left(\frac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$ (D) $\left(\frac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

MCN092

22. Let $A = \left\{ \frac{1967 + 1686i \sin \theta}{7 - 3i \cos \theta} : \theta \in \mathbb{R} \right\}$. If A contains exactly one positive integer n , then the value of n is

[JEE (Advanced) 2023]

MCN093

23. Let z be complex number satisfying $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$, where $\bar{z}$ denotes the complex conjugate of z . Let the imaginary part of z be nonzero.

[JEE (Advanced) 2023]

Match each entry in **List-I** to the correct entries in **List-II**.

List-I	List-II
(P) $ z ^2$ is equal to	(1) 12
(Q) $ z - \bar{z} ^2$ is equal to	(2) 4
(R) $ z ^2 + z + \bar{z} ^2$ is equal to	(3) 8
(S) $ z + 1 ^2$ is equal to	(4) 10
	(5) 7

The correct option is :

- (A) (P) $\rightarrow$ (1) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
 (B) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (3) (S) $\rightarrow$ (5)
 (C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (5) (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)

MCN094

24. Let $A_1, A_2, A_3, \dots, A_8$ be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let PA_i denote the distance between the points P and A_i for $i = 1, 2, \dots, 8$. If P varies over the circle, then the maximum value of the product $PA_1 \cdot PA_2 \cdot \dots \cdot PA_8$, is

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MCN095

25. Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If $\alpha_1, \alpha_2, \alpha_3$, and α_4 are all the roots of the equation $f(x) = 0$, then $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ is equal to

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MCN230

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	C	D	B	C	B	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	A	B	B	B	A	C	C	C
Que.	21	22	23	24	25					
Ans.	D	A	C	A	B					

EXERCISE - O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C,D	A,B	B,C,D	A,B,C,D	B,D	A,B,C,D	B,C,D	B,C	A,B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	B,C	B,D	A,B,C	A,B,C,D	B	C	C	A,B	B,D
Que.	21	22	23	24	25					
Ans.	5.00	3.00	-1.50	B	$A \rightarrow Q,R ; B \rightarrow P,S ; C \rightarrow Q,S ; D \rightarrow P,R$					

EXERCISE - S

1. 99 2. 7 3. 13 4. 25 5. 1 6. 1.41 or 1.42
 7. 1.25 8. 1.50 9. 0.27 or 0.28 10. 6.66 or 6.67

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	4	3	310	4	1	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	13	1	4	3	3	3	2	80	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	4	2	1	14	2	3	1	221
Que.	31	32	33	34	35					
Ans.	2	4	20	49	3					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6				
Ans.	B,C,D	C,D	C	B	C	$A \rightarrow P,Q; B \rightarrow P,Q; C \rightarrow P,Q,S,T; D \rightarrow Q,T$				
Que.	7	8	9	10	11	12	13	14	15	16
Ans.	4	1	A,C,D	A,D	A,B,D	A,C,D	B	3.00	B,C	8.00
Que.	17	18	19	20	21	22	23	24	25	
Ans.	C	B,D	0.50	4.00	A	281	B	512	20	